



```
In [26]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
In [28]: df = pd.read_excel("StudentDb.xlsx")
print(df)
```

	Student_Id	Student_Name	Gender	Email_Id \
0	2301204001	Jadhav Sonali	Female	sonalijadhav@gmail.com
1	2301204002	Hase Ishwari	Female	ishwarihase@gmail.com
2	2301204003	Kadam Dhanraj	Male	dhanrajkadam@gmail.com
3	2301204004	Gunjal Tejal	Female	tejalgunjal@gmail.com
4	2301204005	Hingane Aditya	Male	adityahingane@gmail.com
5	2301204006	Kadatan Shreya	Female	shreyakadatan@gmail.com
6	2301204007	Jadhav Jay	Male	jayjadhav@gmail.com
7	2301204008	Gunjal Tanuja	Female	tanujagunjal@gmail.com
8	2301204009	Ghegadamal Mohit	Male	mohitghegadamal@gmail.com
9	2301204010	Kadalag Rutuja	Female	rutujakadalag@gmail.com

	Mobile_No	DSA	JAVA	Python	DBMS
0	9356547399	80.0	91.0	95.0	70.0
1	8010561032	87.0	96.0	96.0	NaN
2	9430242451	90.0	53.0	89.0	89.0
3	7030386643	45.0	NaN	68.0	87.0
4	9104568920	98.0	84.0	98.0	NaN
5	8967453560	NaN	75.0	76.0	99.0
6	9105445376	90.0	52.0	78.0	67.0
7	7034678790	66.0	63.0	NaN	89.0
8	7082404010	87.0	65.0	84.0	44.0
9	4586952415	NaN	73.0	49.0	67.0

```
In [30]: df.notnull()
```

```
Out[30]:
```

	Student_Id	Student_Name	Gender	Email_Id	Mobile_No	DSA	JAVA	Python
0	True	True	True	True	True	True	True	True
1	True	True	True	True	True	True	True	True
2	True	True	True	True	True	True	True	True
3	True	True	True	True	True	True	False	True
4	True	True	True	True	True	True	True	True
5	True	True	True	True	True	False	True	True
6	True	True	True	True	True	True	True	True
7	True	True	True	True	True	True	True	False
8	True	True	True	True	True	True	True	True
9	True	True	True	True	True	False	True	True

```
In [32]: series1 = pd.notnull(df["DSA"])
df[series1]
```

```
Out[32]:
```

	Student_Id	Student_Name	Gender	Email_Id	Mobile_No
0	2301204001	Jadhav Sonali	Female	sonalijadhav@gmail.com	9356547399
1	2301204002	Hase Ishwari	Female	ishwarihase@gmail.com	8010561032
2	2301204003	Kadam Dhanraj	Male	dhanrajkadam@gmail.com	9430242451
3	2301204004	Gunjal Tejal	Female	tejalgunjal@gmail.com	7030386643
4	2301204005	Hingane Aditya	Male	adityahingane@gmail.com	9104568920
6	2301204007	Jadhav Jay	Male	jayjadhav@gmail.com	9105445376
7	2301204008	Gunjal Tanuja	Female	tanujagunjal@gmail.com	7034678790
8	2301204009	Ghegadamal Mohit	Male	mohitghegadamal@gmail.com	7082404010

```
In [35]: series1 = pd.notnull(df["JAVA"])
df[series1]
```

```
Out[35]:
```

	Student_Id	Student_Name	Gender	Email_Id	Mobile_No
0	2301204001	Jadhav Sonali	Female	sonalijadhav@gmail.com	9356547399
1	2301204002	Hase Ishwari	Female	ishwarihase@gmail.com	8010561032
2	2301204003	Kadam Dhanraj	Male	dhanrajkadam@gmail.com	9430242451
4	2301204005	Hingane Aditya	Male	adityahingane@gmail.com	9104568920
5	2301204006	Kadatan Shreya	Female	shreyakadatan@gmail.com	8967453560
6	2301204007	Jadhav Jay	Male	jayjadhav@gmail.com	9105445376
7	2301204008	Gunjal Tanuja	Female	tanujagunjal@gmail.com	7034678790
8	2301204009	Ghegadamal Mohit	Male	mohitghegadamal@gmail.com	7082404010
9	2301204010	Kadalag Rutuja	Female	rutujakadalag@gmail.com	4586952415

```
In [37]: series1 = pd.notnull(df["Python"])
df[series1]
```

Out[37]:

	Student_Id	Student_Name	Gender	Email_Id	Mobile_No
0	2301204001	Jadhav Sonali	Female	sonalijadhav@gmail.com	9356547399
1	2301204002	Hase Ishwari	Female	ishwarihase@gmail.com	8010561032
2	2301204003	Kadam Dhanraj	Male	dhanrajkadam@gmail.com	9430242451
3	2301204004	Gunjal Tejal	Female	tejalgunjal@gmail.com	7030386643
4	2301204005	Hingane Aditya	Male	adityahingane@gmail.com	9104568920
5	2301204006	Kadatan Shreya	Female	shreyakadatan@gmail.com	8967453560
6	2301204007	Jadhav Jay	Male	jayjadhav@gmail.com	9105445376
8	2301204009	Ghegadamal Mohit	Male	mohitghegadamal@gmail.com	7082404010
9	2301204010	Kadalag Rutuja	Female	rutujakadalag@gmail.com	4586952415

```
In [39]: series1 = pd.notnull(df["DBMS"])
df[series1]
```

Out[39]:

	Student_Id	Student_Name	Gender	Email_Id	Mobile_No
0	2301204001	Jadhav Sonali	Female	sonalijadhav@gmail.com	9356547399
2	2301204003	Kadam Dhanraj	Male	dhanrajkadam@gmail.com	9430242451
3	2301204004	Gunjal Tejal	Female	tejalgunjal@gmail.com	7030386643
5	2301204006	Kadatan Shreya	Female	shreyakadatan@gmail.com	8967453560
6	2301204007	Jadhav Jay	Male	jayjadhav@gmail.com	9105445376
7	2301204008	Gunjal Tanuja	Female	tanujagunjal@gmail.com	7034678790
8	2301204009	Ghegadamal Mohit	Male	mohitghegadamal@gmail.com	7082404010
9	2301204010	Kadalag Rutuja	Female	rutujakadalag@gmail.com	4586952415

```
In [41]: df['DSA']=df['DSA'].fillna(56)
print(df)
```

	Student_Id	Student_Name	Gender	Email_Id \
0	2301204001	Jadhav Sonali	Female	sonalijadhav@gmail.com
1	2301204002	Hase Ishwari	Female	ishwarihase@gmail.com
2	2301204003	Kadam Dhanraj	Male	dhanrajkadam@gmail.com
3	2301204004	Gunjal Tejal	Female	tejalgunjal@gmail.com
4	2301204005	Hingane Aditya	Male	adityahingane@gmail.com
5	2301204006	Kadatan Shreya	Female	shreyakadatan@gmail.com
6	2301204007	Jadhav Jay	Male	jayjadhav@gmail.com
7	2301204008	Gunjal Tanuja	Female	tanujagunjal@gmail.com
8	2301204009	Ghegadamal Mohit	Male	mohitghegadamal@gmail.com
9	2301204010	Kadalag Rutuja	Female	rutujakadalag@gmail.com

	Mobile_No	DSA	JAVA	Python	DBMS
0	9356547399	80.0	91.0	95.0	70.0
1	8010561032	87.0	96.0	96.0	NaN
2	9430242451	90.0	53.0	89.0	89.0
3	7030386643	45.0	NaN	68.0	87.0
4	9104568920	98.0	84.0	98.0	NaN
5	8967453560	56.0	75.0	76.0	99.0
6	9105445376	90.0	52.0	78.0	67.0
7	7034678790	66.0	63.0	NaN	89.0
8	7082404010	87.0	65.0	84.0	44.0
9	4586952415	56.0	73.0	49.0	67.0

```
In [43]: df['JAVA']=df['JAVA'].replace(np.nan,78)
print(df)
```

	Student_Id	Student_Name	Gender	Email_Id \
0	2301204001	Jadhav Sonali	Female	sonalijadhav@gmail.com
1	2301204002	Hase Ishwari	Female	ishwarihase@gmail.com
2	2301204003	Kadam Dhanraj	Male	dhanrajkadam@gmail.com
3	2301204004	Gunjal Tejal	Female	tejalgunjal@gmail.com
4	2301204005	Hingane Aditya	Male	adityahingane@gmail.com
5	2301204006	Kadatan Shreya	Female	shreyakadatan@gmail.com
6	2301204007	Jadhav Jay	Male	jayjadhav@gmail.com
7	2301204008	Gunjal Tanuja	Female	tanujagunjal@gmail.com
8	2301204009	Ghegadamal Mohit	Male	mohitghegadamal@gmail.com
9	2301204010	Kadalag Rutuja	Female	rutujakadalag@gmail.com

	Mobile_No	DSA	JAVA	Python	DBMS
0	9356547399	80.0	91.0	95.0	70.0
1	8010561032	87.0	96.0	96.0	NaN
2	9430242451	90.0	53.0	89.0	89.0
3	7030386643	45.0	78.0	68.0	87.0
4	9104568920	98.0	84.0	98.0	NaN
5	8967453560	56.0	75.0	76.0	99.0
6	9105445376	90.0	52.0	78.0	67.0
7	7034678790	66.0	63.0	NaN	89.0
8	7082404010	87.0	65.0	84.0	44.0
9	4586952415	56.0	73.0	49.0	67.0

```
In [45]: df['Python']=df['Python'].fillna(56)
print(df)
```

	Student_Id	Student_Name	Gender	Email_Id \
0	2301204001	Jadhav Sonali	Female	sonalijadhav@gmail.com
1	2301204002	Hase Ishwari	Female	ishwarihase@gmail.com
2	2301204003	Kadam Dhanraj	Male	dhanrajkadam@gmail.com
3	2301204004	Gunjal Tejal	Female	tejalgunjal@gmail.com
4	2301204005	Hingane Aditya	Male	adityahingane@gmail.com
5	2301204006	Kadatan Shreya	Female	shreyakadatan@gmail.com
6	2301204007	Jadhav Jay	Male	jayjadhav@gmail.com
7	2301204008	Gunjal Tanuja	Female	tanujagunjal@gmail.com
8	2301204009	Ghegadamal Mohit	Male	mohitghegadamal@gmail.com
9	2301204010	Kadalag Rutuja	Female	rutujakadalag@gmail.com

	Mobile_No	DSA	JAVA	Python	DBMS
0	9356547399	80.0	91.0	95.0	70.0
1	8010561032	87.0	96.0	96.0	NaN
2	9430242451	90.0	53.0	89.0	89.0
3	7030386643	45.0	78.0	68.0	87.0
4	9104568920	98.0	84.0	98.0	NaN
5	8967453560	56.0	75.0	76.0	99.0
6	9105445376	90.0	52.0	78.0	67.0
7	7034678790	66.0	63.0	56.0	89.0
8	7082404010	87.0	65.0	84.0	44.0
9	4586952415	56.0	73.0	49.0	67.0

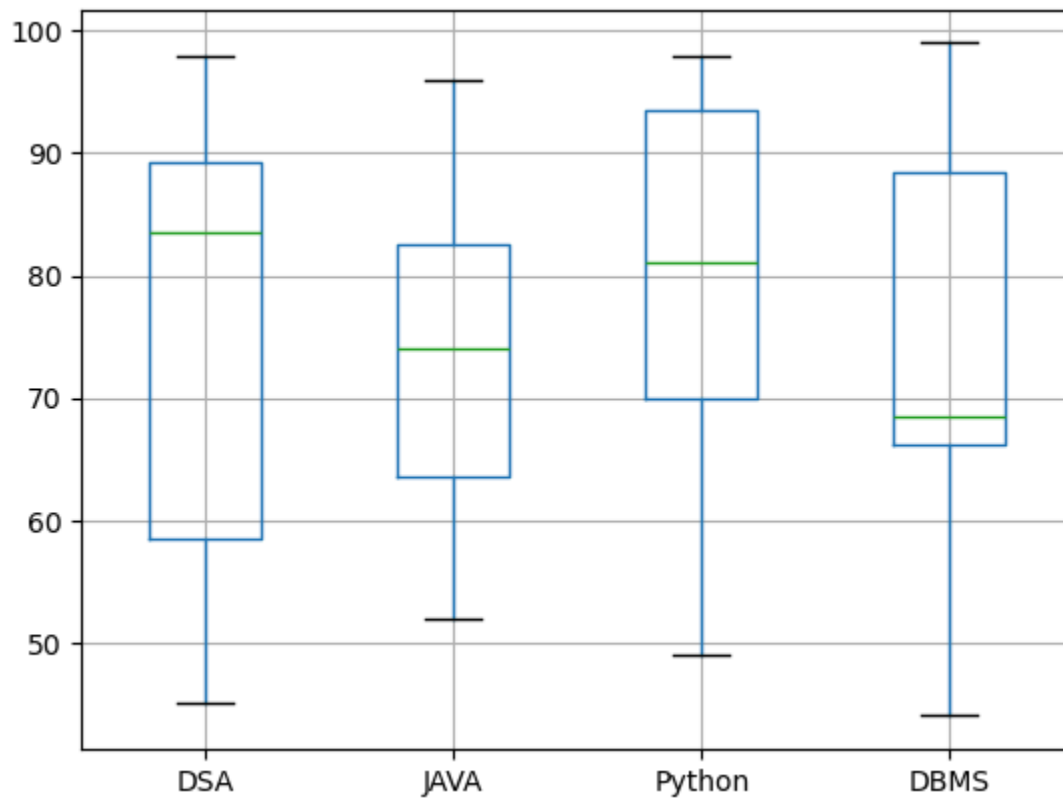
```
In [47]: df['DBMS']=df['DBMS'].fillna(66)
print(df)
```

	Student_Id	Student_Name	Gender	Email_Id \
0	2301204001	Jadhav Sonali	Female	sonalijadhav@gmail.com
1	2301204002	Hase Ishwari	Female	ishwarihase@gmail.com
2	2301204003	Kadam Dhanraj	Male	dhanrajkadam@gmail.com
3	2301204004	Gunjal Tejal	Female	tejalgunjal@gmail.com
4	2301204005	Hingane Aditya	Male	adityahingane@gmail.com
5	2301204006	Kadatan Shreya	Female	shreyakadatan@gmail.com
6	2301204007	Jadhav Jay	Male	jayjadhav@gmail.com
7	2301204008	Gunjal Tanuja	Female	tanujagunjal@gmail.com
8	2301204009	Ghegadamal Mohit	Male	mohitghegadamal@gmail.com
9	2301204010	Kadalag Rutuja	Female	rutujakadalag@gmail.com

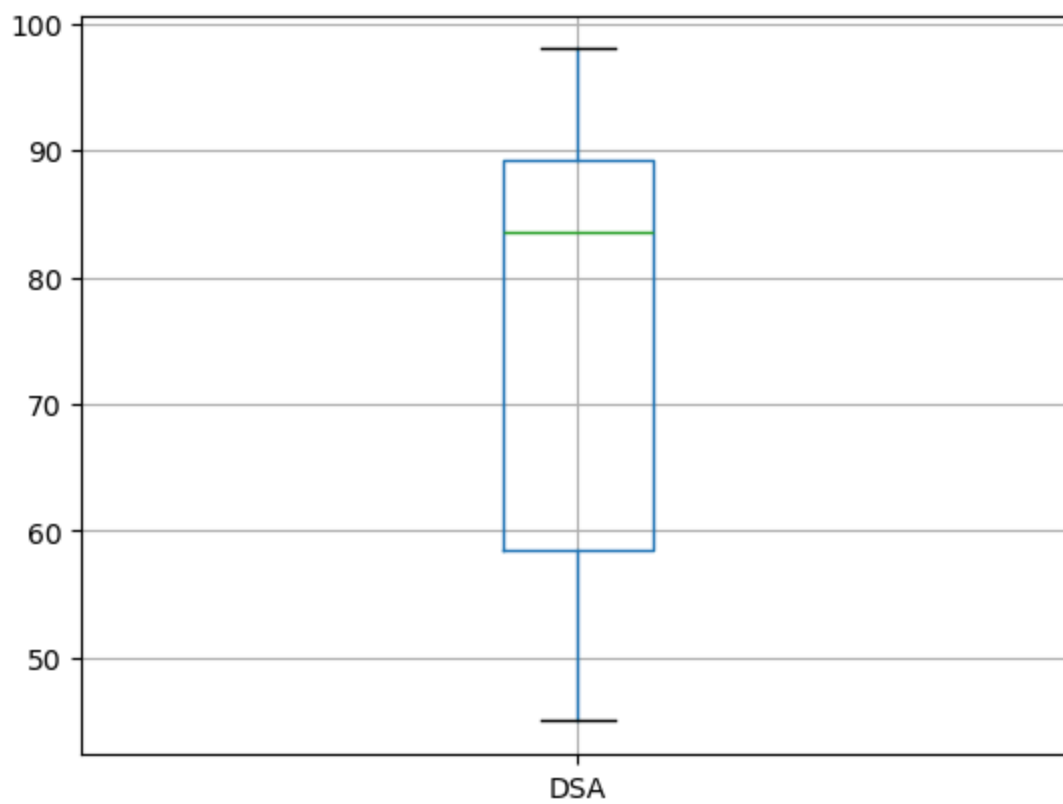
	Mobile_No	DSA	JAVA	Python	DBMS
0	9356547399	80.0	91.0	95.0	70.0
1	8010561032	87.0	96.0	96.0	66.0
2	9430242451	90.0	53.0	89.0	89.0
3	7030386643	45.0	78.0	68.0	87.0
4	9104568920	98.0	84.0	98.0	66.0
5	8967453560	56.0	75.0	76.0	99.0
6	9105445376	90.0	52.0	78.0	67.0
7	7034678790	66.0	63.0	56.0	89.0
8	7082404010	87.0	65.0	84.0	44.0
9	4586952415	56.0	73.0	49.0	67.0

```
In [ ]: col = ["DSA", "JAVA", "Python", "DBMS"]
df.boxplot(col)
```

```
plt.show()
```



```
In [61]: col = ['DSA']  
df.boxplot(col)  
plt.show()
```



```
In [ ]: df['JAVA'].plot(kind = 'hist')  
plt.show()
```

